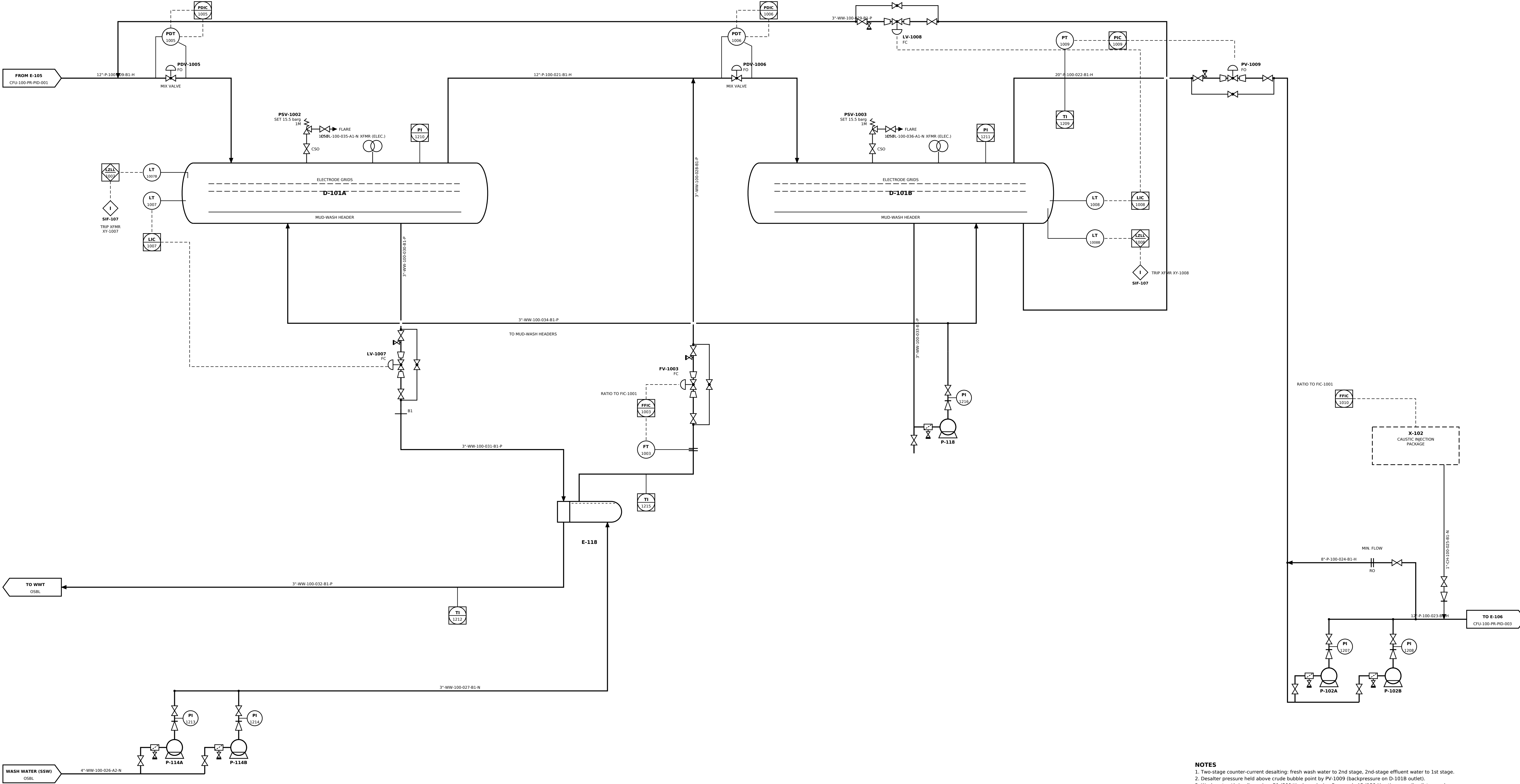


D-101A Electrostatic desalter, 1st stage ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids DESIGN 17.0 barg / 165 C, CA 3 mm MAT'L CS	D-101B Electrostatic desalter, 2nd stage ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids DESIGN 17.0 barg / 165 C, CA 3 mm MAT'L CS	E-118 Desalter wash water / brine 84 m2, 1 shell(s) DESIGN 20 / 20 barg / 160 C, CA 3 mm DUTY 1.62 MW SHELL: Wash water / TUBE: Brine MAT'L CS shell / duplex 2205 tubes	P-102A/B Desalted crude booster RATED 805 m3/h @ 263 m, dP 20 bar MOTOR 719 kW, API 682 Plan 11/52 DES. T 165 C, MAT'L S-6 (CS/12Cr)	P-114A/B Desalter wash water RATED 37 m3/h @ 82 m, dP 8 bar MOTOR 18.3 kW, API 682 Plan 11/52 DES. T 80 C, MAT'L S-6 (CS/12Cr)	P-118 Desalter mud-wash / recycle RATED 19 m3/h @ 64 m, dP 6 bar MOTOR 7.5 kW, API 682 Plan 11/52 DES. T 150 C, MAT'L S-6 (CS/12Cr)	X-102 Caustic injection package (desalted crude) Tank 2 m3 + 2 x 100 % metering pumps MOTOR 0.75 kW MAT'L SS316
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- NOTES**
- Two-stage counter-current desalting; fresh wash water to 2nd stage, 2nd-stage effluent water to 1st stage.
 - Desalter pressure held above crude bubble point by PV-1009 (backpressure on D-101B outlet).
 - Wash water / brine lines class B1 (300#) where design P exceeds A2 (150#) rating - see line list remarks.
 - Transformer trip on low-low interface (SIF-107) by Electrical (XY-1007 / XY-1008 to feeder CB).
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS			
DISCIPLINE		STAGE	SCALE
PROCESS		FEED	NTS
			3 OF 17
DRAWN		CHECKED	DATE
CLAUDE CODE		-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-002			A